
CS246: Database Management Systems Lab

Week # 09 (1 Questions, 90 Marks)

Lab session: AL1

Held on: 10-Mar-2025 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 4

Submission time: 16:45 Hrs, 10-Mar-2025

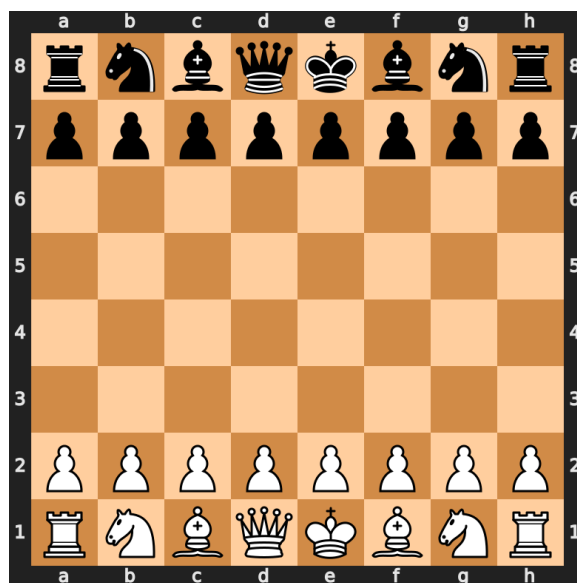
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Important information:

1. Write all the SQL statements in a text file named with `your-roll-number.sql` file name & extension and upload. Replace the text `your-roll-number` with appropriate roll number.
2. Refer to regular expressions
3. Refer to string functions
4. Refer to create procedure
5. Refer to create function



Chess move notation in algebraic form is given below

Chessboard Coordinates : The board is an 8×8 grid

1. Files/Columns are labeled a to h from left to right from White's perspective

2. Ranks/Rows are labeled 1 to 8 from bottom to top from White's perspective
3. The board description is as given below pictorially.

Piece Abbreviations : Following are the important abbreviations

1. K: King
2. Q: Queen
3. R: Rook
4. B: Bishop
5. N: Knight (to avoid confusion with King)
6. Pawns are not given a letter; just the destination square is recorded.

Move Notation : The general move notation is given as described below

1. Each move is written as a combination of the piece moved and the destination square
2. e4: Pawn moves to e4.
3. Nf3: Knight moves to f3.

Special Symbol - Captures : Representation

1. A capture is denoted by an x.
2. Nxf3 means a Knight captures a piece on f3.
3. For pawns, the column it comes from is included before the x.
4. exd5 means a pawn from the e-column captures a piece on d5.

Special Symbol - Check : Representation

1. A check is denoted by +.
2. Qh5+ means the Queen moves to h5 and puts the King in check.

Special Symbol - Checkmate : Representation

1. A checkmate is denoted by #.
2. Qh5#

Special Symbol - Castling : Representation

1. 0-0: Kingside castling (short).
2. 0-0-0: Queenside castling (long).

Special Symbol - Pawn Promotion : Representation

1. When a pawn reaches the last rank, it is promoted. The promotion piece is written after the move.

2. e8=Q means a pawn moves to e8 and becomes a Queen

Special Symbol - En Passant : Representation

1. Denoted as a normal capture, but it's a special pawn move.
2. exd6 e.p.

Special Symbol - Ambiguity : Resolution

1. If two pieces of the same type can move to the same square, the file or rank of the moving piece is included to clarify.
2. Example: If both Knights can move to f3:
 - Nbd2 (Knight from the b-column moves to d2).
 - N1f3 (Knight from row 1 moves to f3).

Result : Representation

1. The game result is written at the end:
2. 1 - 0: White wins
3. 0 - 1: Black wins
4. $\frac{1}{2} - \frac{1}{2}$: Draw

Examples Some examples

1. e4 e5: Meaning
 - White starts by moving the pawn to e4.
 - Black replies with the pawn to e5
2. Nf3 Nc6: Meaning
 - White moves the Knight to f3.
 - Black replies with the pawn to e5

Question 1: (90 points)

Given the above description about chess notation, perform the following tasks.

Task 01 (1 mark) Create a database named **week09**. Create the following tables within this database.

Task 02 (2 marks) Create a table with the following schema

1 st column	First player move	fixed length char of size 6
2 nd column	Second player move	fixed length char of size 6

Task 03 (2 marks) Load the data into the above table.

Task 04 (25 marks) Write SQL procedure to find incorrect knight move. The procedure do not take any input parameters. The procedure must state each knight move as "correct" or "incorrect" and print.

Task 05 (30 marks) Write SQL procedure to find incorrect (black/white) rook move. Move is valid if (1) rook moves only in a row or in a column and (2) there is no obstruction while performing the move. The procedure do not take any input parameters. That is the procedure must state each rook move as “correct” or “incorrect” and print.

Task 06 (30 marks) Write SQL procedure to find incorrect (black/white) bishop move. Move is valid if (1) bishop moves in any diagonal direction and (2) there is no obstruction while performing the move. The procedure do not take any input parameters. That is the procedure must state each bishop move as “correct” or “incorrect” and print.

In the performing the above, you have to use regular expressions in the queries. For any additional information you may create functions and obtain relevant information. You may also use builtin string functions